

Outline of the design research required for phase 2 of the Web Witchcraft and Wizardry project

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24 February 2026

Abstract

An overview of the design research required for phase 2 of the Web Witchcraft and Wizardry project and beyond, along with a report on the manipulated images submitted alongside this essay.

1 Introduction

This short essay outlines the design strategy for the Web Witchcraft and Wizardry project and includes a brief report on the manipulated images submitted alongside the essay. The Web Witchcraft and Wizardry project is a website that aims to educate children about web development using a metaphor of witchcraft and wizardry. This is a continuing redesign of an existing web site and this phase of the redesign focuses on the visual and aesthetic aspects of the site, along with introducing interaction and automation to the site.

2 Report on the submitted manipulated images

Details of how the manipulated images were created are included in the web site hosting them, <https://web.witchcraft-and-wizardry.school/manipulated-images>, so are not included here. The first three images will be used in the redesigned web site while the other three are included to showcase my competence in using digital image manipulation tools and techniques. They have been chosen to demonstrate use of a range of open source tools (LaTeX, Inkscape and Krita) and a variety of techniques (programmatic image generation, tracing, compositing, colour manipulation and typography selection). While more time would have allowed some of the images to be more polished (particularly Shot Morgana), I am happy with the overall quality of the images and the range of techniques demonstrated.

3 Design Goals

The areas for further research are broken down into the four categories covered in the lectures: layout, colour, typography and animation. A search using Google Scholar has been conducted and for each section key papers and books have been identified for review.

3.1 Layout

The main challenge the web site presents is how to present a large amount of text in a way that is engaging and easy to navigate for children. It would be easy to use large blocks of text,

which is what the old site does, but this is not only uninteresting but for many students it will be difficult to capture and maintain their attention. Use of video has been ruled out for now as I don't have the time or skills to edit video content in the quantity required, but this may be considered in the future. Text can be broken up with images but this also presents a challenge in generating sufficient images of a high enough quality to be engaging. Ultimately I may resort to using AI generated artwork, but my experiments thus far have not been promising and I would like to avoid this if possible.

Key concepts in this area to research include the use of whitespace, visual hierarchy and the use of images to break up text and create visual interest. I will also look at how to use layout to create a sense of progression through the site, guiding users through the content in a way that is easy to follow. Lupton and Phillips (2008), Mullet and Sano (1995), Roth et al. (2013), Michailidou et al. (2008), and Williams (2015) are all key books and papers that cover these concepts and will be reviewed in detail.

Another area of research which I find fascinating is the use of eye tracking to understand how users interact with web pages and how this can inform layout decisions. This is currently outside the scope of phase 2 of the project but is something I would like to explore in the future. Coincidentally Holmquist and Anderson (2017) is a comprehensive guide to eye tracking methods and techniques and recently appeared in my Amazon recommendations.

3.2 Colour

After layout, the next most important aspect of the design is colour. The current site uses a very limited colour palette (essentially black on white), and was not designed with either colour theory or accessibility in mind. My intention is for the new design to support both light and dark themes and will be based around two main colours (purple and cyan) used in the Embodied Mind logo. The design work includes creating a colour palette based on these two main colours, and ensuring that the colours used have sufficient contrast to be accessible to users with visual impairments. I will also look at how to use colour to create a sense of hierarchy and guide users through the content. Although web design is modern, the principles of colour go back centuries and there is a wealth of literature on the subject. For example: Lupton and Phillips (2008). Chevreul (1839), Goethe (1810), Gage (1993) and Munsell (1905) all cover colour theory while Elliot and Maier (2007) explores the psychological effects of colour and Cyr et al. (2010) examines cultural differences in colour perception. Noiwan and Norcio (2008), Bonnardel et al. (2011), Hill and Scharff (2003) relate colour to web design and of course the WCAG 2.2 guidelines (Caldwell et al., 2024) remain relevant.

3.3 Typography

Typography, while important, is of less concern than layout and colour in this project. I have already chosen three font families to use (Cinzel for headings and titles, Open Sans for body text and Fira Mono for code snippets) and these will be used in the new design. The main focus in this area will be on ensuring that the typography is consistent throughout the site and that it supports readability and accessibility. Where typography becomes more relevant is in its interaction with the other elements of the design, particularly colour and layout. For example, I will need to ensure that the typography works well with the colour palette and that it is used in a way that supports the visual hierarchy of the site. Key papers in this area include Lupton and Phillips (2008), Bringhurst (2013), Garrett (2010) and McCandless (2012).

3.4 Animation

My main design goal with respect to animation is to use it sparingly and to not use JavaScript where CSS animations will suffice. With modern CSS capabilities, there is no need to use JavaScript for what should be style rather than interaction. My main source of design and technique in this area are the YouTube tutorials on CSS by Kevin Powell (Powell, 2026) which have served me very well so far in my journey through CSS.

Conclusion

The Web Witchcraft and Wizardry redesign project is a large undertaking and there are many areas to research and explore. Phase 1 covered accessibility and usability and phase 2 will focus on the visual and aesthetic aspects of the site, along with introducing interaction and automation. The design strategy outlined in this essay will likely extend beyond the phase 2 delivery of the project in April, but it will provide me with a clear roadmap for the work that needs to be done and the areas that I need to research after this course is completed.

Word count

Word count: 1055 words (excluding references)

References

- Nathalie Bonnardel, Annie Piolat, and Ludovic Le Bigot. The impact of colour on website appeal and users' cognitive processes. *Displays*, 32(2):69–80, 2011.
- Robert Bringhurst. *The Elements of Typographic Style*. Hartley & Marks Publishers, Vancouver, BC, 4th edition, 2013.
- Ben Caldwell, Michael Cooper, Loretta Guarino Reid, and Gregg Vanderheiden. Web content accessibility guidelines (wcag) 2.2. W3c recommendation, World Wide Web Consortium (W3C), 2024. URL <https://www.w3.org/TR/WCAG22/>. Accessed: November 2025.
- Michel Eugène Chevreul. *The Principles of Harmony and Contrast of Colours*. Longman, Orme, Brown, Green and Longmans, London, 1839.
- Dianne Cyr, Milena Head, and Hector Larios. Colour appeals in website design within and across cultures: A multi-method evaluation. *International Journal of Human-Computer Studies*, 68(1-2):1–21, 2010.
- Andrew J Elliot and Markus A Maier. Color and psychological functioning. *Current Directions in Psychological Science*, 16(5):250–254, 2007.
- John Gage. *Colour and Culture: Practice and Meaning from Antiquity to Abstraction*. Thames and Hudson, London, 1993.
- Jesse James Garrett. *The Elements of User Experience: User-Centered Design for the Web and Beyond*. New Riders, Berkeley, CA, 2nd edition, 2010.
- Johann Wolfgang von Goethe. *Theory of Colours*. John Murray, London, 1810. Translated by Charles Lock Eastlake, 1840.

- Alex Hill and Lauren Scharff. Colour and contrast in website design. *Proceedings of Human Factors and Ergonomics Society Annual Meeting*, 47(11):1368–1372, 2003.
- Kenneth Holmquist and Richard Anderson. *Eye tracking: A comprehensive guide to methods, paradigms, and measures*. CreateSpace Independent Publishing Platform, UK, 2017.
- Ellen Lupton and Jennifer Cole Phillips. *Graphic Design: The New Basics*. Princeton Architectural Press, New York, 2008. Covers fundamental principles of layout including scale, hierarchy, layers, transparency, modularity, and grid systems.
- David McCandless. *Information is Beautiful*. Collins, London, 2012. Includes discussion of typography in data visualisation and web design.
- Eleni Michailidou, Simon Harper, and Sean Bechhofer. Visual complexity and aesthetic perception of web pages. *Proceedings of the 26th Annual ACM International Conference on Design of Communication*, pages 215–224, 2008. doi: 10.1145/1456536.1456581.
- Kevin Mullet and Darrell Sano. *Designing Visual Interfaces: Communication Oriented Techniques*. SunSoft Press, Englewood Cliffs, NJ, 1995. Foundational work on visual organization, elegance, and communication through interface layout.
- Albert H Munsell. *A Color Notation*. G.H. Ellis Company, Boston, 1905.
- Jiraporn Noiwan and Anthony F Norcio. The effects of color schemes and banner animation on attention and memory in multimedia advertising. In *Proceedings of the 10th international conference on Human computer interaction with mobile devices and services*, pages 373–376, 2008.
- Kevin Powell. Kevin powell youtube channel, 2026. URL <https://www.youtube.com/@KevinPowell>. Accessed April 2026.
- Silvia P. Roth, Peter Schmutz, Sandrine L. Pauwels, Javier A. Bargas-Avila, and Klaus Opwis. The effect of webpage layout on user engagement with web content. *Interacting with Computers*, 25(5):402–414, 2013. doi: 10.1093/iwc/iwt024.
- Robin Williams. *The Non-Designer’s Design Book*. Peachpit Press, Berkeley, CA, 4th edition, 2015. Accessible introduction to layout principles: contrast, repetition, alignment, and proximity.